1. Scientific/Technical/Management

1.1 Introduction

1.1.1 Motivation

This project aims to establish baseline principles of sediment transportation through a comparative analysis of bedforms across planetary bodies where surface processes occur without biospheric influences. Sedimentary bedforms have been identified on a wide range of Solar System bodies, including Mars, Venus, Titan, and Comet 67P/Churyumov-Gerasimenki (67P/C-G). ***Despite this widespread distribution, the physics controlling bedform onset, morphology, and evolution remains poorly constrained across planetary environments.*** We will address this gap by mapping bedform occurrence, morphology, and orientation on Mars, Venus, Titan, and Comet 67P/C-G and testing whether transport models reproduce the observed patterns under each body's boundary conditions. Previous models have applied a uniform structure across planetary bodies leading to inaccuracies. In order to understand the governing physics behind sediment transportation across planetary surfaces, we will create a map of the distribution and morphology of bedforms present on the target bodies and create a model to reflect the observed patterns and properties under the conditions of each planet. ***We will isolate the environmental variables such as gravity, atmospheric density, and sediment availability across planetary bodies to test whether universal sediment-transport behavior can be distinguished from environment-specific effects.*** We will accomplish this work by combining GIS-based mapping of bedform distributions with environment-specific mobilization thresholds from adapted transport modeling and then compare the results across targets. Results will improve our understanding of when and where regolith is mobile and how active surface processes shape bodies experiencing various conditions throughout the Solar System.

1.1.2 Perceived Impact

This project will produce a standardized cross-planetary bedform dataset that enables direct comparisons of bedform distribution, morphology, and orientation across targets. ***Because dune and ripple orientations reflect dominant transport directions, these products provide a physically grounded way to infer modern wind regimes and sediment availability from remote sensing data.*** By integrating planetary geomorphology with transport modeling, we will identify which physical variables most strongly control sediment mobility across bodies with differing atmospheres, gravities, and sediment compositions. For comet 67P/C–G, this approach also provides a pathway to evaluate how transient gas flows associated with outgassing may generate ripple-like bedforms despite the absence of a sustained atmosphere[5].

Methodologically, this project combines remote sensing with computational modeling of aeolian transport processes. By coupling geomorphological mapping with numerical simulations, we will establish a framework linking observed landforms to the conditions under which they formed. This could be applied to future missions, such as Dragonfly or DAVINCI, and provide valuable context for data analysis and mission planning. Isolating physical principles governing sediment transport will enhance the understanding of soil formation and erosion on Earth, where human activity increasingly alters natural sediment balances.

The anticipated outcomes of this work include ***(1) global GIS layers for each target that map bedform occurrence and dominant orientation***; ***(2) modeled mobility thresholds (e.g., threshold shear velocity and predicted regions of likely transport) tailored to each body's environmental parameters***; and (***3) a comparative synthesis that identifies universal versus***

environment-specific controls on sediment transport, providing context for mission planning and landing-site evaluation.

1.1.3 Relevance to NASA and the Solicitation

Aligned with NASA Solar System Science priorities, this project will focus on (1) analyzing existing planetary mission datasets to create new reference maps to address, "*The analysis of planetary data [...] generation of new reference information, such as laboratory measurements; and creation of geologic maps*" and (2) developing workflows/software to improve dataset usability for future investigations addressing, "*The development and dissemination of software tools that facilitate the use of existing datasets, or that would enable or enhance future science investigations.*" Additionally, this work connects with the 2023-2032 Decadal Survey by answering question 5.6: "*What drives active processes occurring in the interiors and on the surfaces of solid bodies*?"

1.2 Background

1.2.1 Sediment Transport Across Planetary Bodies

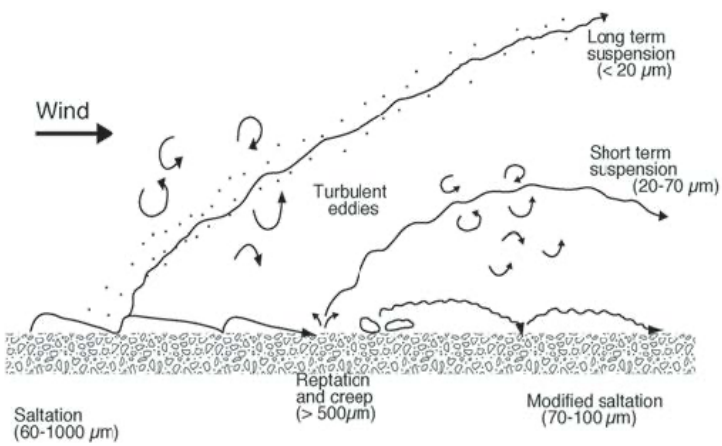

Figure 1. Illustration depicting aeolian sediment transportation. This figure highlights grain-size dependency of different modes, including surface creep, saltation, and suspension. Particle trajectories and turbulent eddies illustrate how fluid force redistributes sediments, leading to the formation of bedforms. *Adapted from the U.S. National Park Service*[3].

Most dunes on Earth form through aeolian transportation, in which wind moves grains by first overcoming the physical forces that hold them in place. ***Entrainment is the initiation of motion, when a surface grain is dislodged and set into movement by wind-driven shear stress.*** Because grains are partially sheltered by neighboring particles, entrainment occurs only when applied shear stress exceeds resisting forces such as gravity and interparticle cohesion. Once mobilized, grains move by surface creep when grains are moved along the ground through collisions with other grains, saltation when grains are temporarily lifted by wind and are moved along the surface in a series of skips, or suspension when grains are lifted entirely into the air and transported for a distance before settling. Collisions during suspension can then entrain additional grains (Figure 1) where the initiation of motion follows a "Goldilocks" principle in which particles that are too small are held by cohesion and those that are too big are restricted by weight.

For terrestrial planets with atmospheres, sand-sized particles are the easiest to mobilize. ***Aeolian transportation requires wind that can supply sufficient shear stress to overcome resisting forces, so sediment motion is sensitive to atmospheric and gravitational conditions***.

Terrestrial planets vary widely in terms of gravity and atmospheric density, which directly influence sediment entrainment thresholds.

For example, Mars has a thin atmosphere composed of carbon dioxide, roughly 1% of Earth's pressure [8, 12], and low gravity, which result in a reduced sediment weight and fluid density. In contrast, Venus has an extremely dense carbon dioxide atmosphere, nearly two magnitudes greater than Earth[13-14], extreme temperatures, and gravitational acceleration similar to Earth, resulting in a lower required shear stress for particle transport. Even among just the terrestrial planets, sediment transportation behaves under widely different boundary conditions.

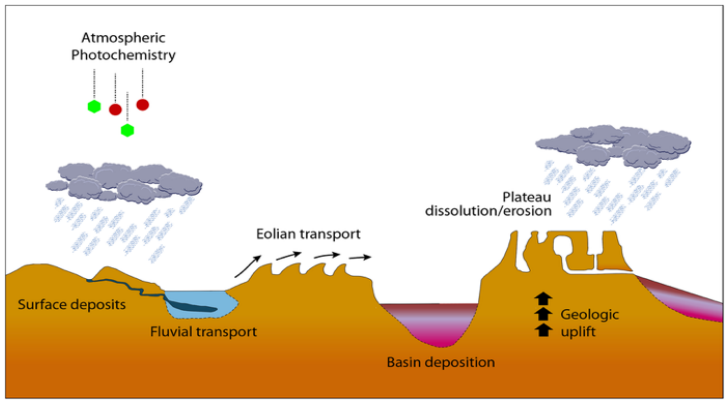

Figure 2. Titan's sediment–karst cycle links atmospheric organic production, transport, and subsurface karst formation. Atmospheric photochemistry generates organic sediments that are redistributed by fluvial and aeolian processes and concentrated in basins. Changes in topography and hydrocarbon rainfall promote dissolution and conduit formation, producing karst landforms. This model frames Titan as a system where sediment transport is closely coupled to surface chemistry and subsurface modification. *Adapted from Wynne et al. (2022)*[15].

These differences become even more pronounced when extending comparisons beyond the terrestrial planets to bodies with different atmospheres, gravities, and sediment compositions. ***This project focuses on distinct sedimentary mechanisms represented by Titan and Comet 67P/C–G and testing sediment-transport physics under their unusual boundary conditions.***

Titan, like the terrestrial planets, has a dense atmosphere capable of moving small grains. However, what makes Titan unique is the arrangement in which sediment production, transport, and surface modification are all closely related. Organic particles are generated in the upper atmosphere through atmospheric photochemistry and settle at the surface, creating an abundance of sediment for transport. Titan's dense atmosphere and reduced gravity create an environment that encourages aeolian and fluvial transport[1]. Rains and heavy winds dominate transport in a mechanical method similar to Earth; however, the methane rainfall drives chemical dissolution and karst formation shown in Figure 2.

Traveling into the Jupiter family comets, comet 67P/C-G provides an intriguing scenario, where, despite lacking a sustained atmosphere, it exhibits ripple-like bedforms[4]. It is hypothesized that Comet 67P/G-C can create aeolian bedforms due to the sublimation of the ice core, creating gas that builds under the dust mantle[4]. This results in lateral vapor flow, which initially is not strong enough to move sediments. ***However, at perihelion, the pressure gradient formed by the rapid cooling in the shaded regions during the rotation of the comet, combined with the surface wind, is enough to create aeolian bedforms[4].***

1.2.2 Previous Modeling of Sediment Transport

Sediment transport models aim to predict when grains begin moving and how transport rates change once motion starts. Recent work by Pähtz and Durán suggests that transport can often be expressed using scaling relationships[11] that remain consistent across a wide range of conditions, including changes in gravity, atmospheric density, and sediment characteristics (Figure 3). ***Although these scaling relationships are promising, applying them to specific planetary environments requires tailoring model inputs to each body's boundary conditions, which is the focus of Task 2 in our timeline.***

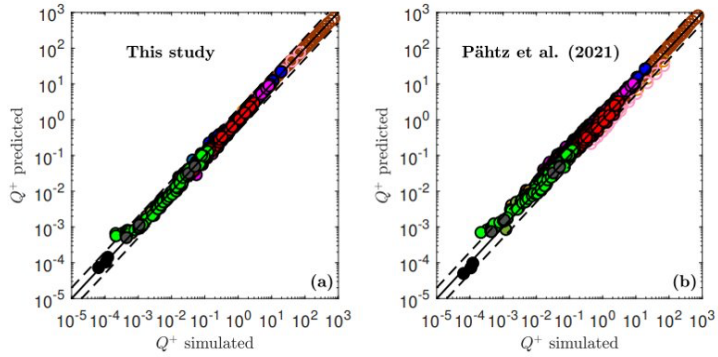

Figure 3. Dimensionless predicted and simulated sediment transport rates across different environments are plotted on a log-log scale. Results show alignment along the 1:1 line, representing strong agreement between the predicted and simulated values. The updated framework in Pahtz's 2023 paper provides increased accuracy when compared to their 2021 results. The consistency across several orders of magnitude provides evidence towards applicability across diverse planetary environments. *Adapted from Pähtz and Durán 2023*[11].

At the same time, real planetary surfaces rarely meet ideal assumptions. Cohesion, cementation, unusual grain compositions (e.g., organics), and limited sediment supply can shift transport thresholds and change how bedforms form and persist. Because of this, models must be tested against observations, which require consistent maps of where dunes and ripples occur (Task 1).

1.3 Proposed Work

To identify the controls on sediment transport across diverse planetary environments, we will integrate remote sensing–based mapping of bedform properties (**Task 1**) with numerical modeling of sediment mobilization and transportation (**Task 2**). These observations will provide quantitative constraints for adapted transport equations and computational fluid dynamics (CFD) simulations that account for each body's gravity, atmospheric density, and material properties. Together, these results will be synthesized into a comparative framework that links sediment sources, sediment availability, transport mechanisms, and depositional patterns across bodies (**Task 3**). This coordinated approach will identify the governing parameters of sediment mobility and distinguish universal behavior from environment specific effects across four bodies.

Task 1: Mapping. Using publicly available spacecraft datasets, we will map bedform occurrence and quantify bedform morphology and orientation using GIS-based analysis. For Mars, we will use Context Camera (CTX) imagery[8] and, where available, High Resolution Imaging Science Experiment (HiRISE)[9] from Mars Reconnaissance Orbiter to measure dune-field extent, crestline orientation, and representative morphology. For Titan, we will use radar backscattering from Cassini[2] to map dune-field distribution and dominant orientations. For Venus, we will use Synthetic Aperture Radar (SAR) from Magellan[5] to identify and map radar-bright bedform patterns where present. For Comet 67P/C–G, we will use OSIRIS images from Rosetta[6] to

identify ripple-like bedforms and measure their spatial distribution and orientations. All mapped features and measurements will be compiled into global body maps for cross-target comparison.

Task 2: Transport Modeling. To interpret the mapped bedforms, we will model the conditions required for sediment mobilization under each body's boundary conditions. We will apply adapted aeolian transport equations and CFD simulations that account for differences in gravity, atmospheric density/composition, and sediment properties. Using an open-source CFD tool (e.g., OpenFOAM), we will test how variations in wind regime and near-surface flow conditions reproduce the observed orientations and morphologies, producing predicted mobility thresholds and spatial patterns of likely transport.

Task 3: Comparative Framework. Finally, we will integrate the mapping and modeling outputs to construct a comparative sediment-transport framework that highlights universal and environment-specific processes. By comparing observed bedform distributions with predicted mobility across bodies, we will identify which parameters most strongly govern sediment mobility under varying gravitational, atmospheric, and compositional conditions.

1.4 Management Plan and Project Timeline

The PI and Co-Is will lead the project while mentoring undergraduates to assist with the mapping of the various planets (Task 1). The PI will also mentor a graduate student that will develop the sediment mobilization model (Task 2). The Co-Is and PI will work together to create the final comparative sediment transportation framework (Task 3). We expect to produce four sediment distribution maps and a peer-reviewed journal article focusing on the updated sediment mobilization model and physical controls. We will also attend two conferences, likely the Lunar and Planetary Science Conference and the American Astronomical Society Division of Planetary Science meeting. Figure 5 shows the proposed timeline for this project.

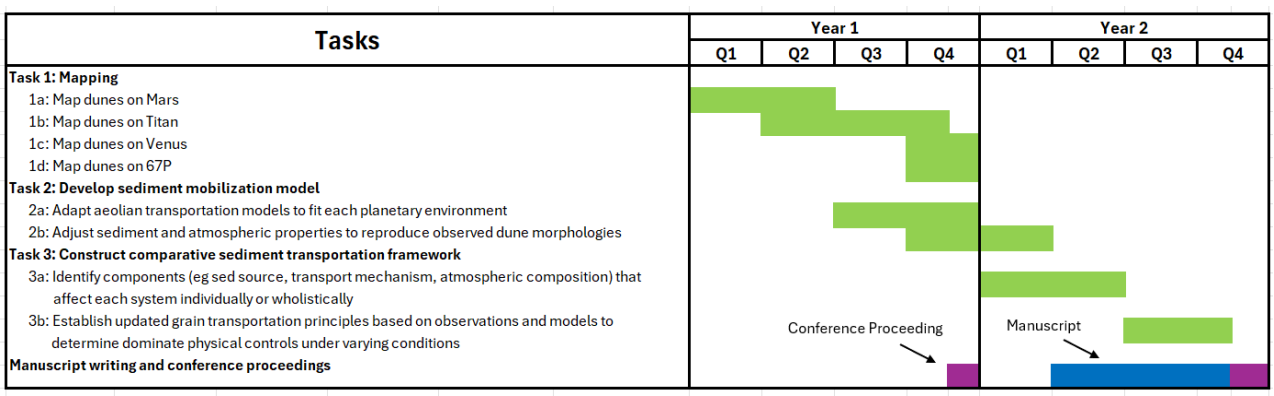

Figure 5. Proposed Timeline of the Project

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